Calculate intervals across zero

Use the scaffold, show each step and ask for an adult check when marked.

Name: ____ Date: __________________

1. Read the model: What is the difference between -4 and 7? Copy or complete the first step.

2. The temperature is -6 C and rises by 9 C. What is it now?

3. How far is -12 from 5 on a number line?

4. Miro says: Miro counts both zero and the starting number when finding the interval from -4 to 7. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: What is the difference between -4 and 7? Copy or complete the first step.
   Answer 11.

2. The temperature is -6 C and rises by 9 C. What is it now?
   Answer 3 C.

3. How far is -12 from 5 on a number line?
   Answer 17.

4. Miro says: Miro counts both zero and the starting number when finding the interval from -4 to 7. Is this correct? Explain.
   Answer Count the jumps between numbers: 4 jumps to zero and 7 more to reach 7, giving 11.

Calculate intervals across zero

Work independently. Show a complete method and check at least one answer.

Name: ______________________________ Date: ____________

1. What is the difference between -4 and 7?

2. The temperature is -6 C and rises by 9 C. What is it now?

3. How far is -12 from 5 on a number line?

4. Solve this independently and show every stage: What is the difference between -4 and 7?

5. Use a second method or representation for: The temperature is -6 C and rises by 9 C. What is it now?

6. Check this result using an inverse, estimate or known fact: How far is -12 from 5 on a number line?

7. What number is halfway between -8 and 6?

8. Identify and correct this misconception: Miro counts both zero and the starting number when finding the interval from -4 to 7.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: How far is -12 from 5 on a number line?

Teacher answers and checking prompts

1. What is the difference between -4 and 7?
   Answer 11.

2. The temperature is -6 C and rises by 9 C. What is it now?
   Answer 3 C.

3. How far is -12 from 5 on a number line?
   Answer 17.

4. Solve this independently and show every stage: What is the difference between -4 and 7?
   Answer 11.

5. Use a second method or representation for: The temperature is -6 C and rises by 9 C. What is it now?
   Answer Expected result: 3 C. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: How far is -12 from 5 on a number line?
   Answer Expected result: 17. A valid check must be shown.

7. What number is halfway between -8 and 6?
   Answer -1.

8. Identify and correct this misconception: Miro counts both zero and the starting number when finding the interval from -4 to 7.
   Answer Count the jumps between numbers: 4 jumps to zero and 7 more to reach 7, giving 11.

9. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: How far is -12 from 5 on a number line?
    Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Calculate intervals across zero

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ____ Date: __________________

1. What number is halfway between -8 and 6?

2. Prove the answer to this question, rather than only calculating it: What is the difference between -4 and 7?

3. Solve this in two different ways and compare them: The temperature is -6 C and rises by 9 C. What is it now?

4. Work backwards from the answer to reconstruct the question: 17.

5. Explain why this reasoning fails: Miro counts both zero and the starting number when finding the interval from -4 to 7.

6. Create a new example that tests the same idea as: How far is -12 from 5 on a number line?

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. What number is halfway between -8 and 6?
   Answer -1.

2. Prove the answer to this question, rather than only calculating it: What is the difference between -4 and 7?
   Answer Expected result: 11. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: The temperature is -6 C and rises by 9 C. What is it now?
   Answer Expected result: 3 C. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: 17.
   Answer One valid reconstruction is: How far is -12 from 5 on a number line? Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro counts both zero and the starting number when finding the interval from -4 to 7.
   Answer Count the jumps between numbers: 4 jumps to zero and 7 more to reach 7, giving 11.

6. Create a new example that tests the same idea as: How far is -12 from 5 on a number line?
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.